Answer Key - Engineering - Graduate Level Assessment 14 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Initial value is 2, slope is -4; B) Initial value is -1, slope is 2; C) Initial value is 1, slope is 2; D) Initial value is 0, slope is undefined; E) Initial value is 0, slope is 2
   Note: Correct option: A - Initial value is 2, slope is -4

2. **Answer:** E
   Options: A) 100 ohms; B) 25 ohms; C) j 100 ohms; D) 50 ohms; E) - j 50 ohms
   Note: Correct option: E - - j 50 ohms

3. **Answer:** A
   Options: A) linearly.; B) cubically.; C) parabolically.; D) logarithmically.; E) hyperbolically.
   Note: Correct option: A - linearly.

4. **Answer:** C
   Options: A) 0; B) 1; C) 2; D) 10
   Note: Correct option: C - 2

5. **Answer:** A
   Options: A) q = hA; B) q = h^2 A; C) q = hA(T^2 - T_0^2); D) q = h / A; E) q = hA/T
   Note: Correct option: A - q = hA

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '8,762.3 psi'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '0'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $e^(1/2)$t cos (1/2) t V. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.60 or 60%, 0.95 or 95%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key